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CH 6: BASIC PROJECT PLANNING TECHNIQUES, PART 1

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Planning Process



The purpose of project planning is to clarify the system requirements and prepare the project plan. The planning process involves answering the following questions, and roughly in this order:

1. What is the desired end result?

Define the project objectives, scope, and system requirements.

2. How will the result be achieved?

Define the work activities, tasks, or jobs to be done to achieve the objectives and requirements.

3. Who will do it?

Specify the project organization—the individuals or departments and contractors, and their *responsibilities*.

Planning Process (cont'd)



4. When and in what order?

Create a schedule showing the timing of work activities, deadlines, and milestone dates.

5. How much?

Create a budget and resource plan to fund and support the project.

6. How well?

Specify a method for *tracking and controlling* project work—for keeping the project conforming to the planned work, schedule, budget, and user and system requirements.

The answers to these and other questions appear in the **project execution plan**.

Common Elements of Project Execution Plan



1. Scope Statement and SOW
2. Detailed requirements
3. Project organization and responsibility for tasks
4. Detailed work definition (WBS or PBS and work package/work task details)
5. Detailed schedules with milestones
6. Project budget and cost accounts
7. Quality plan
8. Risk plan
9. Work review plan
10. Testing plan

Elements of Project Plan, continued



11. Change control plan
12. Documentation plan
13. Procurement plan
14. Implementation plan
15. Health, safety, environment plan
16. Communication plan
17. Financial plan

Scope and SOW



- Scope and/or SOW: The first item on the project execution plan. Two variations of same theme
- Purpose
 - provide broad description of master plan/project to stakeholders
 - directed at core project team, project organization, primary stakeholders

Scope



- Describes “breadth of project,” areas to be covered by project and deliverables & areas not covered.

Includes:

- Objectives of project from perspective of contractor
- Requirements
- Deliverables
- Milestones
- Limits and exclusions: what project does *not* include

Statement of Work—SOW

- SOW is the scope document for *contracted* projects
 - Appears in RFP, proposal, contract, as well as master plan

Defining the SOW

1. For contracted project work

- Contractor and customer agree on definition of work required, work proposed, and basis for costs, schedules, and related matters.
- There are two SOWs,
 - SOW in master plan
 - SOW in *contract* (CSOW)
- SOW in contractor's project plan must contain same information and requirements as stated in CSOW.
- Contractor's SOW and CSOW might be worded differently, but both should have exact same interpretation in terms of work and end results

Defining the SOW

2. Suggestions

- Ensure that SOW and WBS correspond to each other. Both must be clear; neither contractor nor customer question what has to be done.
- Requirements for every end-item, task, and report must be clear enough so parties *responsible* will be able to sign-off acceptance of results.
- Never specify tasks using “as necessary” or “as required”.
 - Where judgments must be made, specify **who** will make them, *procedures* for making them, and potential impact of judgments on cost and schedule escalation.

Issues in Defining SOW



Suggestions:

- Specify requirements using active terminology (“shall” or “will”)
- Never use passive terminology (“should” or “try to”).
- “shall” = must do
- “will” = desirable to do

Issues in Defining SOW



Suggestions

- Categorize specifications applicable to entire project separately from those applicable to only parts of project.
- Hold meetings with customers and technical specialists to review clarity and completeness SOW and WBS.

Work Definition

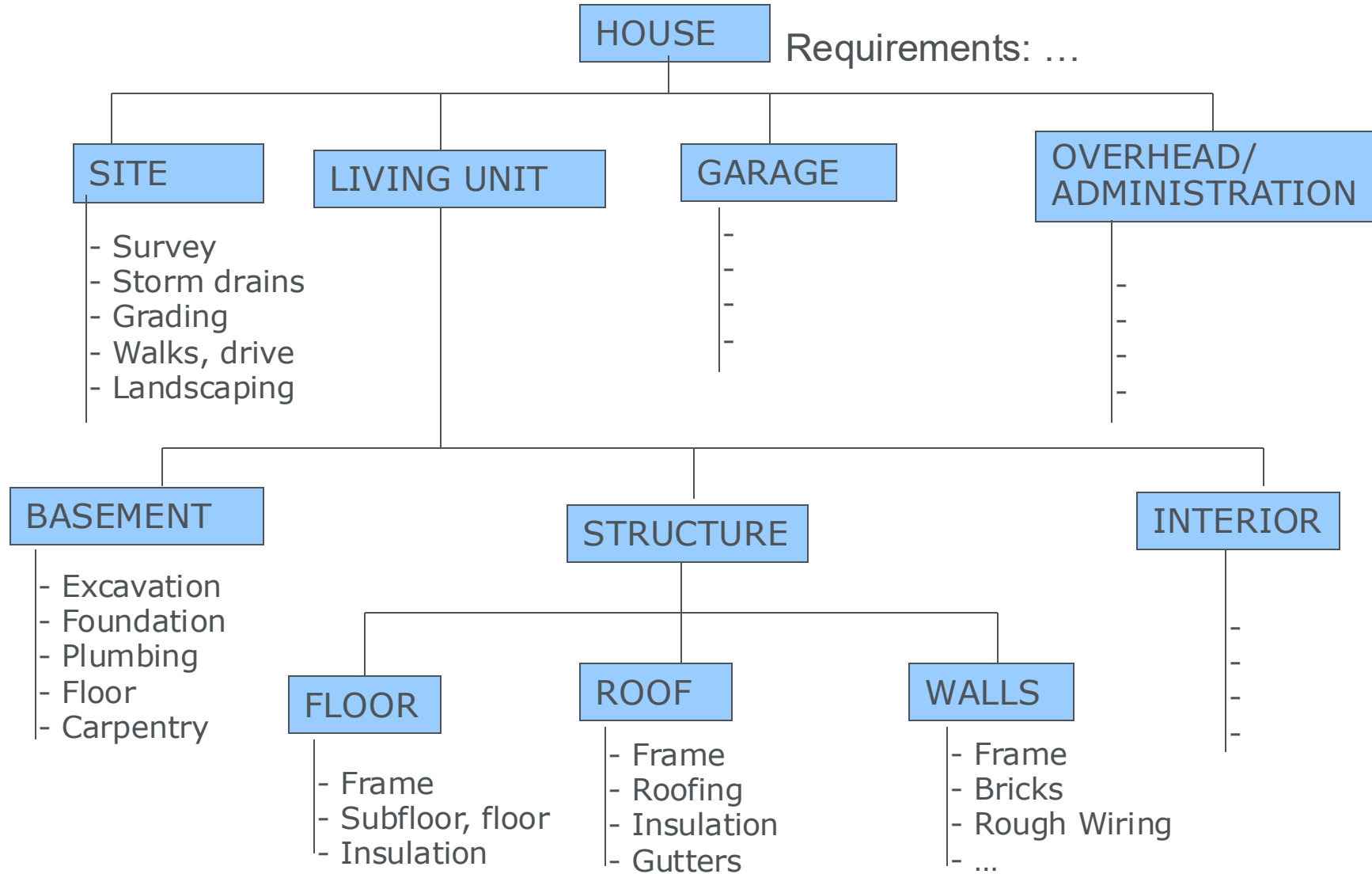


- Start with the SOW and requirements (the “what” of the project)
- Ask “how” will the SOW and requirements be met: what is the actual work to be performed to meet the requirements?
- Subdivide the project into small, well defined work packages
- Use the Work Breakdown Structure, WBS
(a.k.a. Project Breakdown Structure, PBS)

WBS

- Divide project into “well-defined” tasks
- Well-defined tasks: the basis for project schedule, budget, resource requirements, responsibility assignments, and risk management

WBS Example for House



WBS Procedure



- Start with SOW and requirements
- Ask “what 5-10 high level activities would yield intended results?”
- For each high-level activity ask “What is involved here, what is required?”
- Questions that are difficult to answer require the activity to be further broken down
- Continue breaking down activities until all activities at bottom of WBS are **well-defined**

WBS Procedure



- What is well-defined?
 - Well-defined activity is called a “work package” and has following features
- Work Package (ideally) contains:
 - SOW and requirements
 - Clear definition of work and all activities
 - Time estimates or deadlines
 - Cost estimates
 - Responsibility
 - Immediate predecessors, preconditions, inputs
 - Deliverables
 - Resources
 - Risk assessment

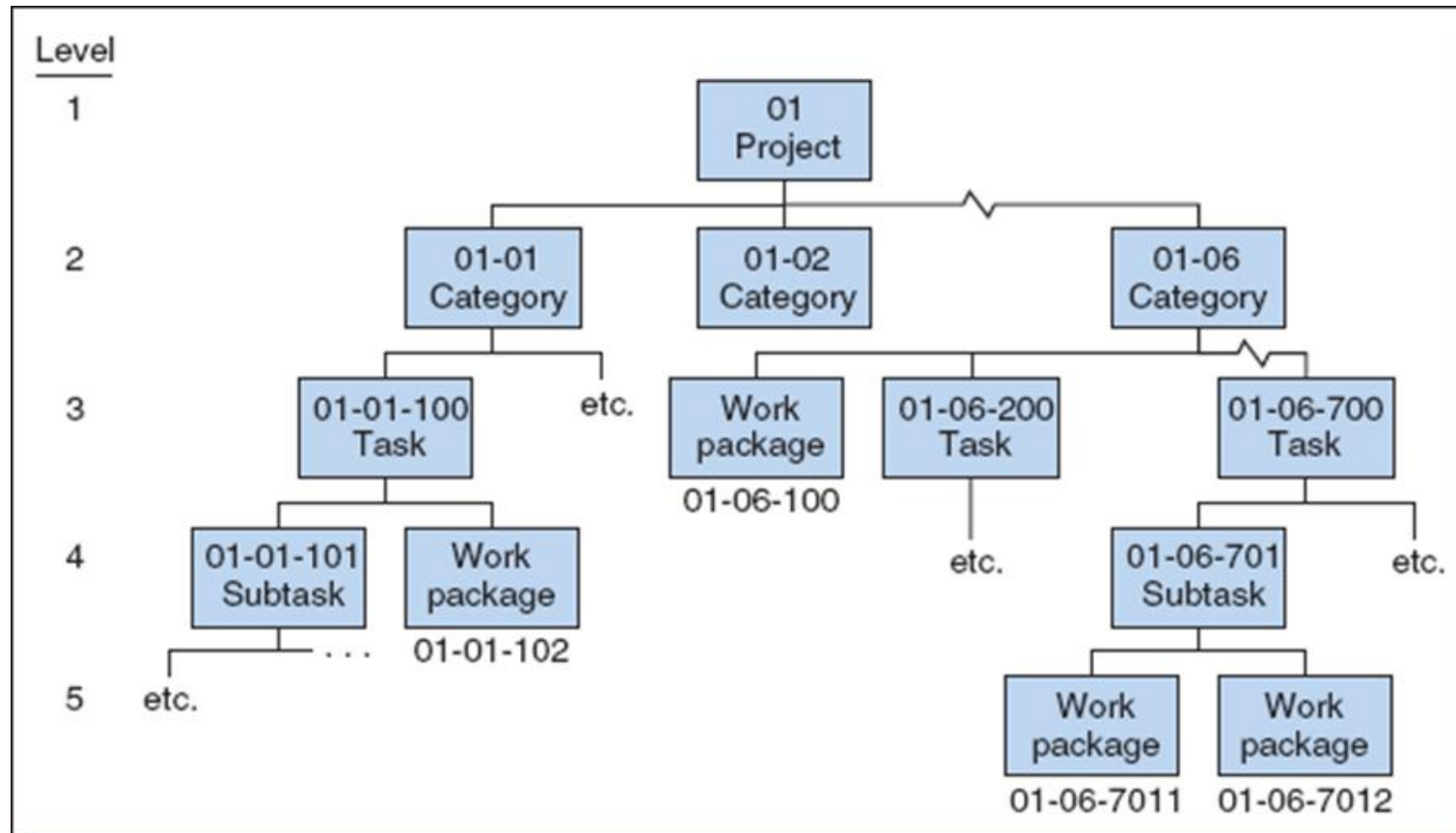
Work Breakdown Structure (WBS)



- WBS divide overall project into major categories, which are then divided into subcategories, etc.
- Examples of levels of categories:

Level	Element Description
1	Project
2	Sub-project or deliverable
3	Work Package
4	Activity

Elements of a WBS, example



Work Packages—end results of WBS



While going through WBS, ask following questions about each work package:

1. *Do you need better estimates of duration and cost of the work package?*
2. *Can you identify who will be responsible for work the package?*
3. *Is the size of work package too large to track and control?*
4. *Are activities within work package independent of each other?*
5. *Do some activities within work package have different immediate predecessors?*
6. *Are risky and non-risky activities combined in the same work package?*
7. *Does the work package contain many different kinds of resources?*

If answer is yes to any of questions, decompose the work package into smaller work packages.

If answer is no to all of them, the work package probably does not need to be subdivided.

Creating WBS

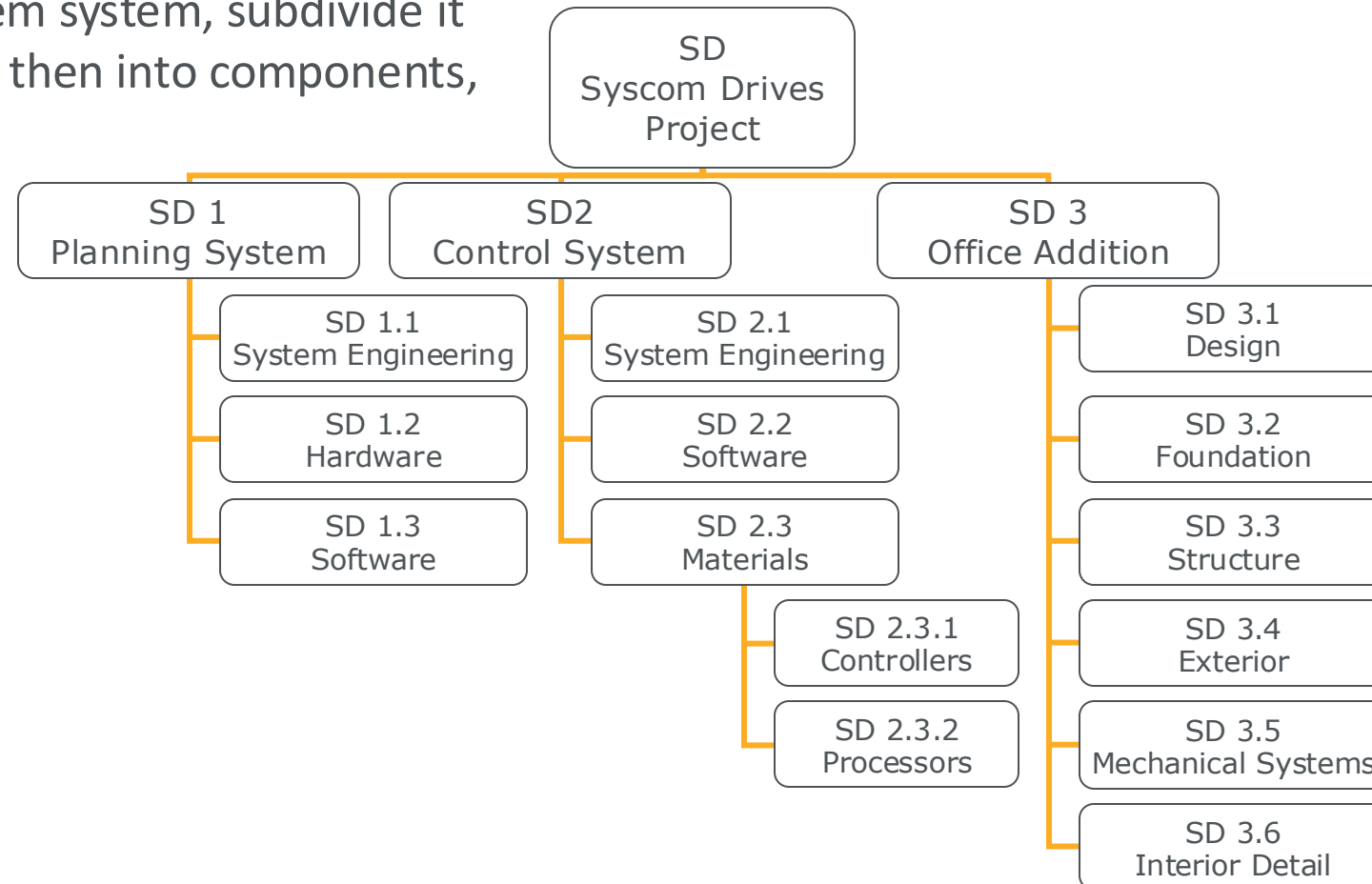


- Project team
 - Brainstorm
 - Past experience
 - Templates
- Multiple teams
- Experts

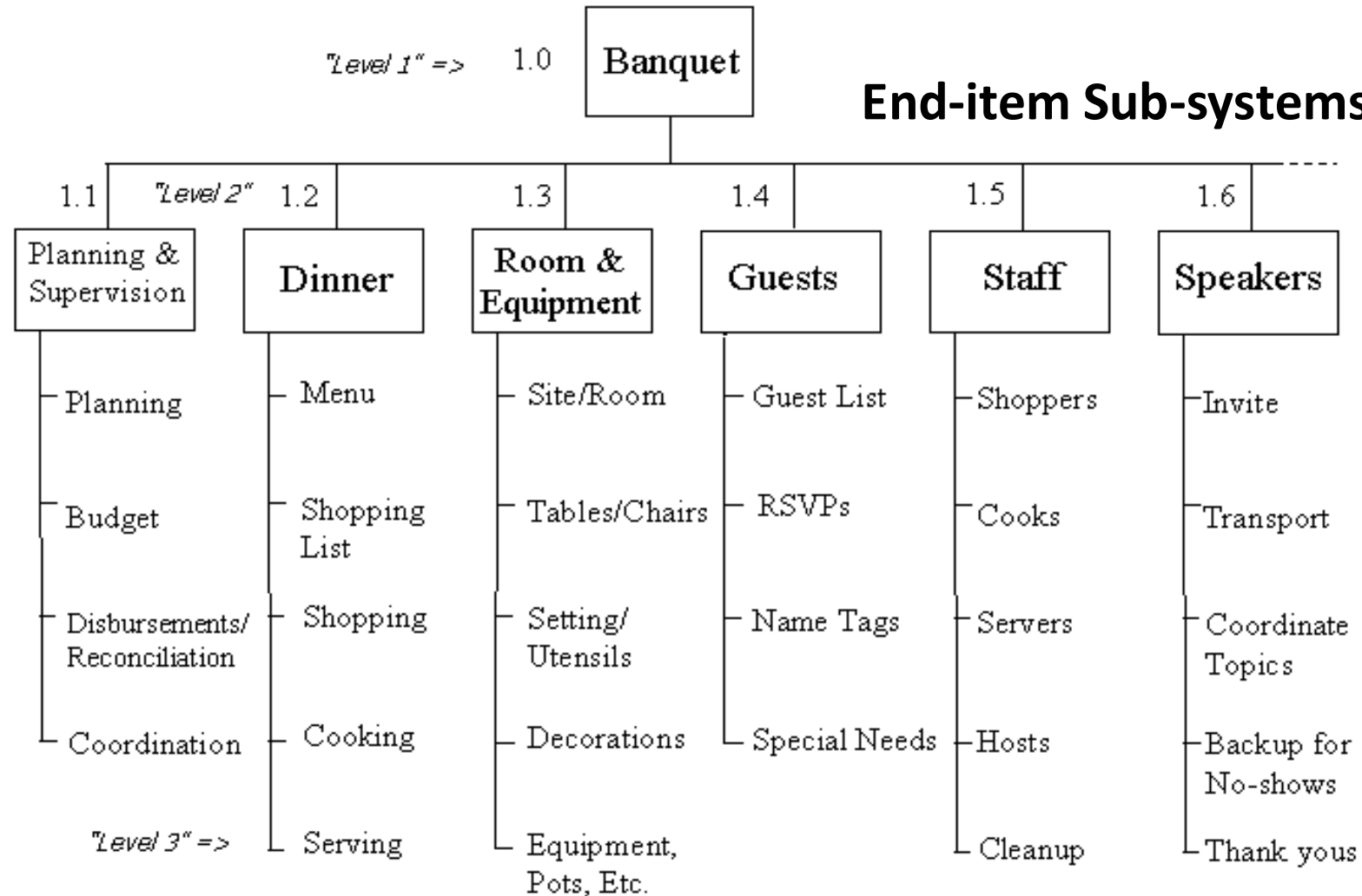
Creating a WBS, Approaches

End-item Sub-systems Approach

Start with the end-item system, subdivide it first into subsystems, then into components, then parts

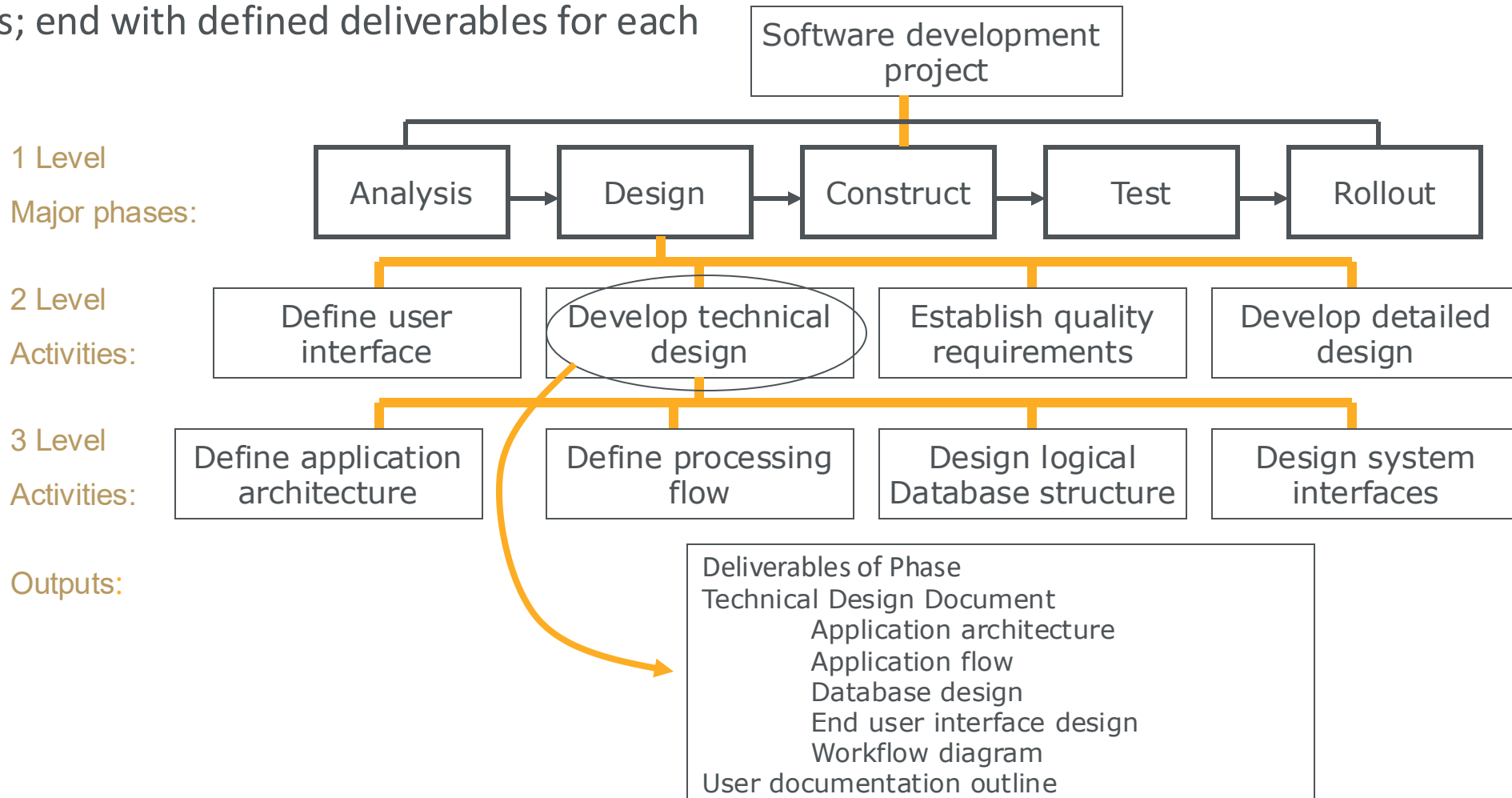


Creating a WBS, Approaches



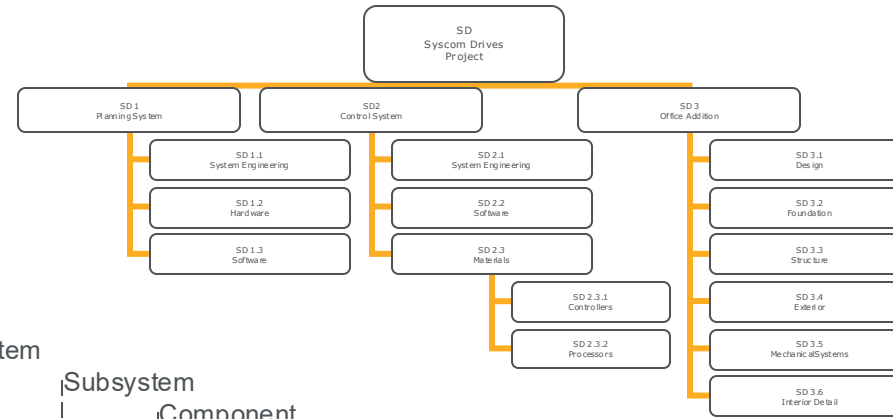
Creating a WBS, Approaches

Process-Steps Approach: Start by defining phases or stages in project, then subdivide each into detailed tasks; end with defined deliverables for each

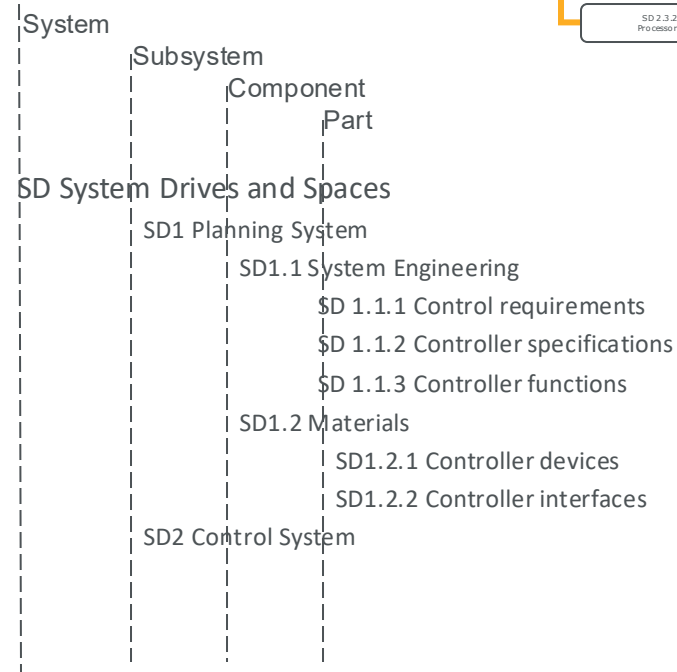


WBS Formats

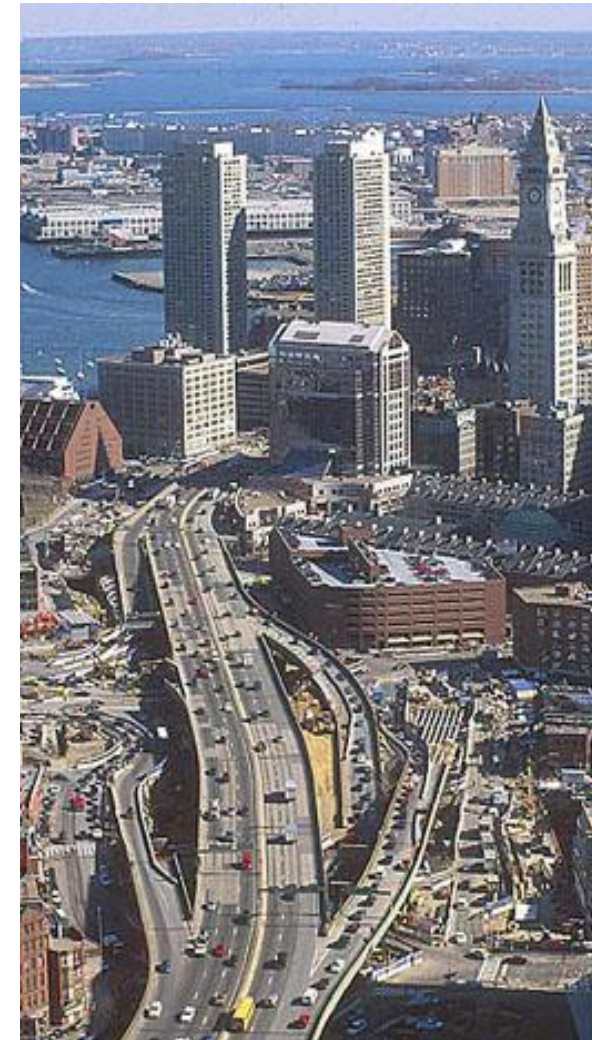
- Tree Structure



- Indented Structure



Boston Big Dig Example

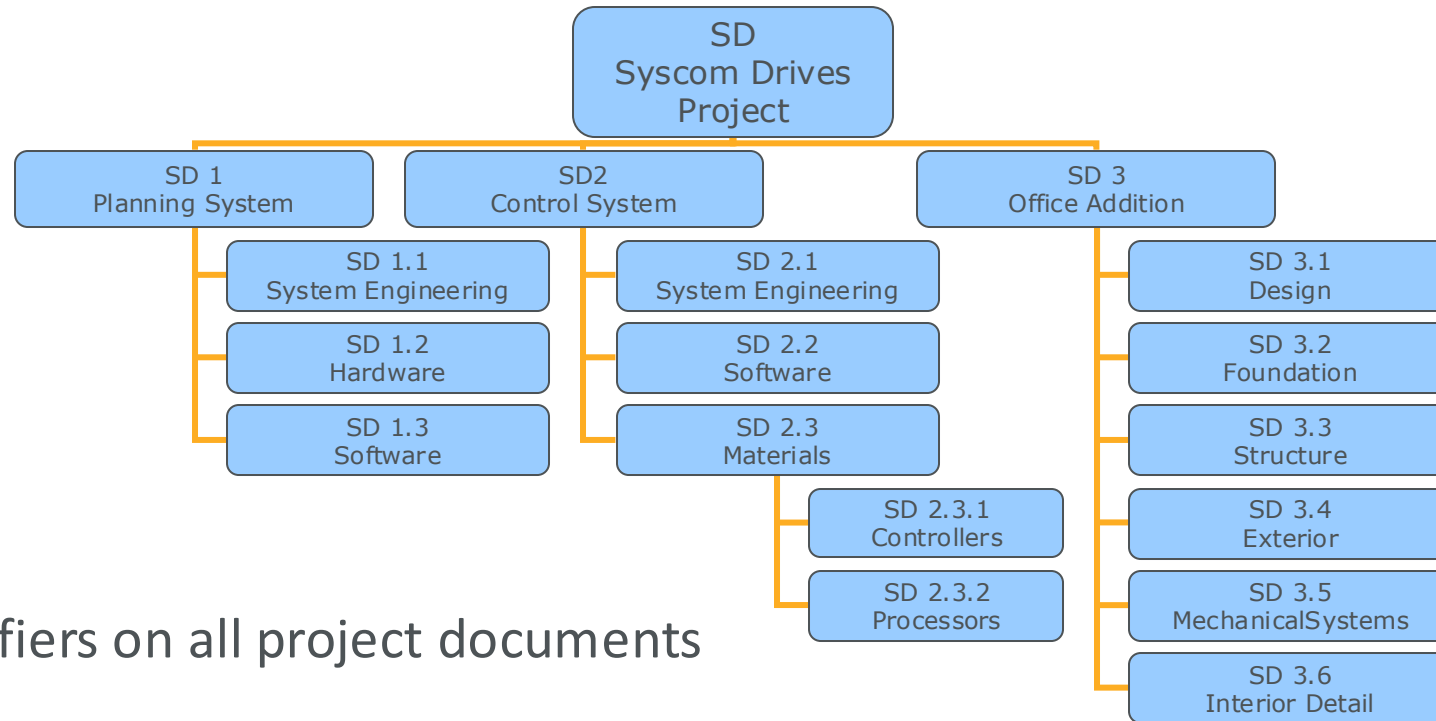


<http://www.cincinnati.com>

- # Downtown Boston



WBS leads to Document Structure



Use same identifiers on all project documents

- Schedules
- Budgets
- Responsibility matrix
- Change requests, etc.

WBS and Integrated Planning and Control



The WBS process and work packages provide integrated plan and control:

1. Functional managers, subcontractors, and others responsible for the work are identified and become involved during the WBS process.
2. Work packages in each phase are logically and physically related to those in earlier and later phases; predecessor requirements are met and no steps overlooked.
3. Work packages are the basis for budgets and schedules. The project budget is the sum of the budgets for all the work packages plus project overhead and indirect expenses. The project schedule is the composite of the schedules of all the work packages.
4. The project organization is formed around work packages. Resources are assembled for and management responsibility delegated to individuals in each work package.
5. The project is directed by directing people working in individual work packages.
6. Project control is exercised through control of work packages.

An integrated project plan is a systems approach to management

Thank you
Enkosi
Dankie

